Stick-together calculations

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find a common final velocity using total momentum.

SUCCESS CRITERIA - I can...

1. Add the initial momenta with signs.
2. Use the combined mass after sticking.
3. Check the result against the direction of total momentum.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “inelastic” and “initial” to explain your model or result. Sentence starters are optional.

inelastic: a collision that transforms some kinetic energy to other forms

initial: at the beginning of the interval; subscript i

final: at the end of the interval; subscript f

01 Warm-up: bring back an earlier idea

Two carts separate from rest with no external horizontal impulse. One has momentum 15 kg·m/s left.
Find the other’s momentum.

Word help: external = acting from outside the chosen system

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v_f = (m_1 v_{1,i} + m_2 v_{2,i})/(m_1 + m_2)$.
Sticking collision in one dimension with negligible external impulse. v_f is the shared final velocity.

Worked reasoning: A 2 kg cart at +3 m/s sticks to a 1 kg cart at rest. Total momentum 6 kg·m/s divided by total mass 3 kg gives final velocity +2 m/s.

Your turn: Mark the conserved quantity on a before-and-after sketch.
Explain why masses are added in the final denominator.

Sentence starter: My claim is … . The diagram/data show …, so … .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Stick-together calculations

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two carts stick.
Identify which masses belong in the final momentum expression and why.

Sentence starter: My claim is … . The diagram/data show …, so … .

Before contact: add velocity arrows.



Marzano check - circle: 1 2 3

04 Apply the physics

A 1 kg cart moves right at 6 m/s and sticks to a 2 kg cart at rest.
Find initial total momentum, combined mass and final velocity. Organize the known quantities on the before-and-after model.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student averages the two starting velocities without considering mass. When would that happen to work?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

Stick-together calculations

antocciphysics

Continue down each column.

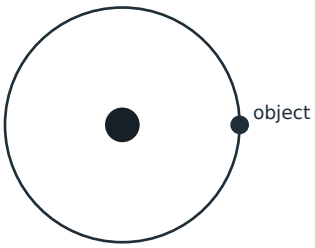
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A car follows a circle at steady 6 m/s.
Sketch velocity at two points. Could net force be zero while it follows this path?
Label velocity and inward acceleration on the circular model.

Word help: net force = the vector sum of all forces on one object

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: 1 2 3

07 Exit check

Two equal 1 kg carts approach at 2 m/s in opposite directions and stick.
Find final velocity. Organize the known quantities on the before-and-after model.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Choose the mass of a stationary cart so a 2 kg cart at 6 m/s slows to 2 m/s when they stick. Organize the known quantities on the before-and-after model.

Sentence starter: I notice ... . This supports ... because ... .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3